

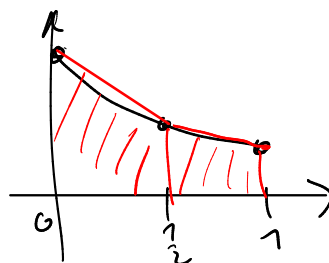
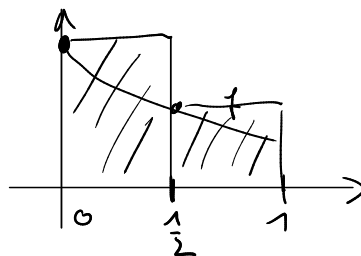
①

$$\int_0^1 \frac{1}{1+x^2} dx$$

a) exakt

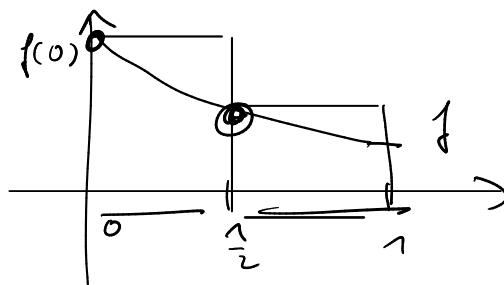
b) Rechteckregel

c) Trapezregel



$$\begin{aligned} a) \quad \int_0^1 \frac{1}{1+x^2} dx &= [\arctan(x)]_0^1 \\ &= \frac{\pi}{4} - 0 \\ &= \frac{\pi}{4} \approx 0,7854 \end{aligned}$$

$$b) \quad \int_0^1 \frac{1}{1+x^2} dx$$

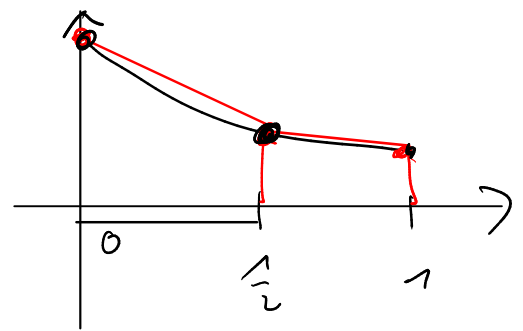


$$\approx \frac{1}{2} f(0) + \frac{1}{2} f\left(\frac{1}{2}\right) =: R$$

$$f(0) = \frac{1}{1+0} = 1, \quad f\left(\frac{1}{2}\right) = \frac{1}{1+\frac{1}{4}} = \frac{4}{5}$$

$$R = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{4}{5} = \frac{1}{2} + \frac{2}{5} = \frac{9}{10} = 0,9$$

c)



$$T = \frac{1}{2} \frac{f(0) + f(\frac{1}{2})}{2} + \frac{1}{2} \frac{f(\frac{1}{2}) + f(1)}{2}$$

$$= \dots = \frac{31}{40} = 0, \underline{775}$$

$$\frac{\pi}{4} \approx 0, \underline{7854}$$